

Granbom, "Convex Polytopes", 2nd ed.

Q. V. Klee; Does every n -polytope have
a realization in $\mathbb{R}^n$ with all integer
vertices?

Clearly true in 2D.

Such in low dim, more "room" to
wiggle vertices so definitely true?

We give counterexample due to M.A. Perles.

Section 1 Gale transform.

Take $X = (x_1, \dots, x_n)$, n points in $\mathbb{R}^d$,
with $\dim \text{aff } X = r$.

Embed in $\mathbb{P}^d$:

$$D_0 = \left(\begin{array}{cccc} x_{1,1} & \dots & x_{1,d} & 1 \\ \vdots & & \vdots & \vdots \\ x_{n,1} & \dots & x_{n,d} & 1 \end{array} \right) \left. \vphantom{\begin{pmatrix} x_{1,1} \\ \vdots \\ x_{n,1} \end{pmatrix}} \right\} n.$$

$\overbrace{\hspace{10em}}^{d+1}$ $\underline{1}$

$\text{Rank } D_0 = r+1,$

so $\exists$ $r+1$ lin. indep. columns including $\underline{1}$.

Defn $M(X) = \text{span}(\text{those columns}) = \text{range } D_0 \subseteq \mathbb{R}^n.$

lemma. $M(X)^\perp = \left\{ a \in \mathbb{R}^n : (a, y) = 0 \ \forall y \in M(X) \right\}$
 $= \left\{ a \in \mathbb{R}^n : \sum_{i=1}^n a_i x_i = 0 \text{ and } \sum_{i=1}^n a_i = 0 \right\}.$

$\dim M(X)^\perp = \text{nullity } D_0 = n - (r+1).$

Interpretation $M(X)^\perp = \left\{ \text{linear functionals that kill all the } x_i \right\}$
 $= \left\{ \text{hyperplanes through all the } x_i \right\}.$

Pick a basis for $M(X)^\perp$ and use it as k columns of a matrix!

$D_1 = \begin{pmatrix} a_{1,1} & \dots & a_{1,n-r-1} \\ \vdots & & \vdots \\ a_{n,1} & \dots & a_{n,n-r-1} \end{pmatrix}$ $\text{codesp. } D_1 = M(X)^\perp.$

$$D_1 = \begin{pmatrix} a_{1,1} & \dots & a_{1,n-r-1} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \dots & a_{n,n-r-1} \end{pmatrix}$$

$$\text{colsp. } D_1 = M(K)^\perp.$$

What is rowsp. D_1 ?

If $x \in \mathbb{R}^n$ lies in $(\text{rowsp. } D_1)^\perp$,

$$\text{then } x^T D_1 = 0$$

$$(x \cdot a_{\cdot, i}) = 0 \quad \forall i$$

ie. x lies on all hyperplanes.

$$\Rightarrow \text{rowsp. } D_1 = \left(\bigcap \left\{ \text{hyperplanes through all } a_i \right\} \right)^\perp.$$

We have a basis D_1 ~~consists~~ $(a_{1,1}, \dots, a_{1,n-r-1}), \dots$
 $(a_{n,1}, \dots, a_{n,n-r-1})$.

Combing D_0 & D_1 rowwise, for each $x_i \in \mathbb{R}^n$ we have $a_i \in \mathbb{R}^{n-t-1}$.

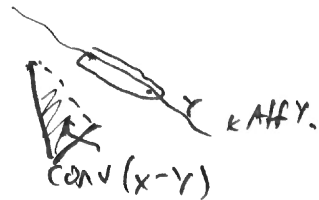
Set $\bar{x}_i = a_i$, $\bar{X} = (\bar{x}_i)$. This is
 a Gale transform of X .

Convex geometry 2 Gale tasks.

Defn. Let $X = (x_i)_{i=1}^n \subseteq \mathbb{R}^n$. Let $Y \subseteq X$.

Say Y is a face of X if

$$\text{conv}(X - Y) \cap \text{Aff } Y = \emptyset.$$



(If $X = P(0)$ for $P = \text{conv } X$ a polytope;

then Y is a face of X iff $\text{conv } X$ is a face of $\text{conv } X$).

Say Y is a coface of X if $X - Y$ is a face.

Theorem. $Y \subseteq X$ is a coface iff either $Y = \emptyset$ or $0 \in \text{relint conv } \bar{Y}$

(here, $\bar{Y}$ means "take the elements of $\bar{X}$ corr. to Y ").

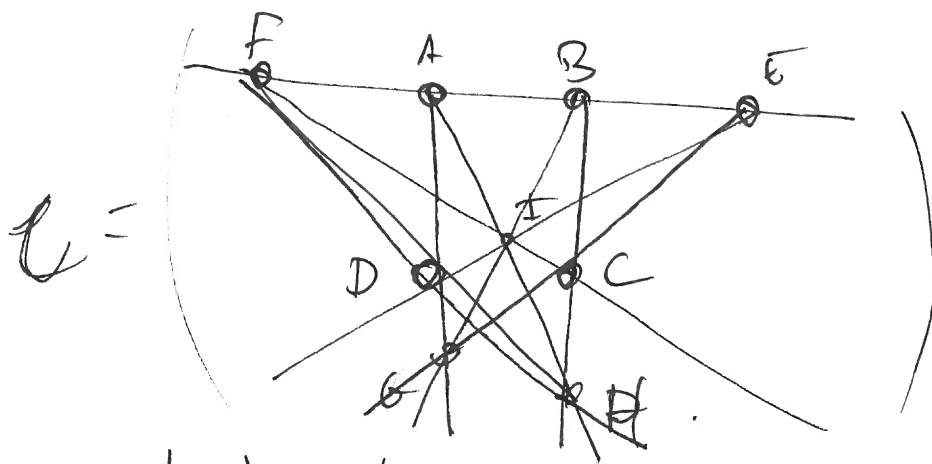
Remark: the idea is to convert affine properties of X to linear properties of $\bar{X}$. So the theorem says the convex property of being a coface translates to spanning a linear subsp. not an affine subsp.

Lemma. If P, P' are polytopes in $\mathbb{R}^d$,
 then P is aff. eq. to P' iff
 their Gale transform of the vertices are
linearly equivalent.

(Anyways.)

Impetus Lemma. If $X \subseteq \mathbb{R}^d$ & aff $X = \mathbb{R}^d$,
 then $X = P(0)$ for some polytope P
 iff either $\bar{x} = 0 \ \forall x \in X$, or
 $\forall$ open half space H^+ of $\mathbb{R}^{n-d-1}$
 satisfies $|\{x \in X: \bar{x}_H \in H^+\}| \geq 2$.

Lemma. Sectn 2. Rigidity!
 Take 9 pts in $\mathbb{P}^2$ with the following
 dependency pattern:



This cannot be resolved with pts having all rational

Proof. The cross-ratio $(A, B; E, F) = \frac{1}{2}(3 + \sqrt{5})$ ^{coords. in $\mathbb{R}^2$.}

Theorem. J agony type P in $\mathbb{R}^8$ with 12 vertices & if no realization of P can have entirely rational vertices.

(So higher dim $\neq$ more flexible!).

Proof. Embed $\mathcal{L} \hookrightarrow \mathbb{R}^3$ in the $z=1$ plane.

$$\text{Set } \hat{V} = \left\{ \frac{A}{\|A\|}, \frac{B}{\|B\|}, \frac{C}{\|C\|}, \dots, \frac{H}{\|H\|}, \frac{-F}{\|F\|}, \frac{-G}{\|G\|}, \frac{H}{\|H\|}, \frac{-I}{\|I\|} \right\}.$$

The saturation lemma for shows it is the Gale diagram of the set of vertices of a 8-poly P with 12 vts.

Suppose P' is some rational rep. of P .

Let $\hat{V}' =$ Gale diagram of $P'(0)$. So a subset of $\hat{V}$ lies on a line/plane through 0 iff the corr. subset of $\hat{V}'$ lies on such a linear subsp.

$\Rightarrow$ the 12 pts of $\hat{V}'$ lie on a line through 0 . In turn with an appropriate plan, get $\mathcal{L}$ back.

If P' were rational, $\hat{V}'$ could be chosen to have rational direction coordinates, then all the 9 pts would be rational, contrary prev. lemma.